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Theory of Carbon NMR Chemical Shifts in Conjugated Molecules

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A molecular orbital theory for carbon-13 chemical shifts in conjugated molecules is formulated. Consideration of the various possible contributions to the shift shows that the local paramagnetic term σ_p^{AA} is expected to dominate. By a detailed LCAO analysis of σ_p^{AA} , its variation with the salient features of the electron environment is determined. It is demonstrated that there is a significant local-charge dependence which agrees in sign and order of magnitude with an experimentally established correlation. In addition, the shielding is shown to be a function of the free valence of the atom under consideration and of the polarity of its sigma bonds. A linearized equation for the carbon-13 chemical shift (with respect to benzene) is presented and compared with the available experimental data.

1. INTRODUCTION

THE chemical shifts of carbon-13 nuclei have been studied experimentally in a variety of conjugated and aromatic systems.¹⁻⁴ One result of these measurements that has aroused considerable interest is the apparent correlation between the carbon-13 shift and the pi-orbital electron density of the atom under consideration. Lauterbur² and Spiesecke and Schneider³ have suggested, on empirical grounds, that the shielding of a carbon atom (A) in a conjugated system varies linearly with the corresponding pi-electron density ρ_A . This leads to an expression for the chemical shift $\Delta\sigma_A$ (with benzene as the reference) of the form

$$\Delta\sigma_A = \alpha(\rho_A - 1), \quad (1)$$

where α is a positive constant that has been shown to equal approximately 160 ppm by comparison of the cyclic systems $C_7H_7^+$, C_6H_6 , $C_6H_5^-$.³ If Eq. (1) is valid, it offers a technique for an "experimental" examination of the details of the charge distribution in appropriate molecules. For example, the results of NMR measurements for azulene and its derivatives have been used with Eq. (1) to "test" pi-electron molecular orbital calculations for these systems.^{2,3}

The extensive possibilities for utilizing a relationship like Eq. (1) suggest that a theoretical examination of its origin should be made and some understanding of its limitations should be provided. In this paper, we determine the various contributions to the carbon-13

chemical shift and show that a significant charge-dependent term of the type given in Eq. (1) is to be expected. However, we find that other molecular properties are also of some importance and attempt to estimate the effect of some of these (e.g., free valence, bond polarity). Comparisons are provided with the available experimental data. Although the agreement is encouraging, it is evident that additional measurements are needed to test the theoretical results.

2. THEORY

The total shielding constant σ for a particular nucleus can be approximated by a sum of three terms⁵:

$$\sigma = \sigma_d + \sigma_p + \sigma', \quad (2)$$

The diamagnetic contribution σ_d is the Lamb term for the atom being considered; i.e., σ_d corresponds to the uniform circulation of the atomic electrons as if they were free. The paramagnetic term σ_p is a local correction due to the molecular environment and involves the mixing of ground and excited electronic states by the applied magnetic field (σ_p would be zero in an isolated atom). The term σ' represents the effect of the circulation on all other atoms and any interatomic ring currents which cannot be localized.

The value of σ_d is given by the well-known formula

$$\sigma_d = (e^2/3mc^2) \sum_i \langle r_i^{-1} \rangle, \quad (3)$$

where $\langle r_i^{-1} \rangle$ is the mean inverse distance of electron i from the nucleus and the summation is over electrons on the atom considered. This gives an immediate linear correlation with electron density of the right sign, but the magnitude of the proportionality constant is much too small as compared with Eq. (1). With Slater

* Alfred P. Sloan Foundation Fellow.

¹ P. C. Lauterbur, J. Chem. Phys. **26**, 217 (1957).

² P. C. Lauterbur, J. Am. Chem. Soc. **83**, 1838, 1846 (1961).

³ H. Spiesecke and W. G. Schneider, Tetrahedron Letters **1961**, 468.

⁴ H. Spiesecke and W. G. Schneider, J. Chem. Phys. **35**, 731 (1961).

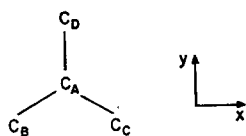


FIG. 1. Cartesian axes for a trigonal carbon and three neighbors.

atomic orbitals, the increase in σ_d if a $2p$ electron is added is 14 ppm, only about 10% of the observed effect. If account is taken of the expansion of the orbitals as the total density is increased, this would be even less. Clearly σ_d cannot make the principal contribution to the charge dependence. The effect of circulations on other atoms are also very small and do not contribute shifts of more than a few parts per million. The main contribution to the carbon-13 chemical shift must therefore originate in changes in the "local paramagnetic" term σ_p . Saika and Slichter⁵ came to this conclusion for fluorine some years ago and made a satisfactory order-of-magnitude calculation. An extension of their work to fluorine in aromatic systems has been made by Karplus and Das.⁶ More recently, Pople has re-examined the theory and applied it to the susceptibility and shielding of a number of systems.⁷

For the estimation of σ_p , it is convenient^{6,7} to use LCAO molecular orbital theory, with each molecular orbital Ψ_i expressed as a linear combination of valence-shell atomic orbitals ϕ_μ ; i.e.,

$$\Psi_i = \sum_{\mu} c_{i\mu} \phi_{\mu}. \quad (4)$$

For carbon atoms the set ϕ_μ contains the $2s$ and all three $2p$ atomic orbitals, while for hydrogen only the $1s$ function is included. If the molecule is placed in a magnetic field, the LCAO coefficients $c_{i\mu}$ are changed slightly by amounts which can be calculated by perturbation theory. By examining the local magnetic field at the nuclear position, it is possible⁷ to deduce a formula for the tensor σ_p^{AA} , the local paramagnetic contribution to the shielding of atom A. For $(\sigma_p^{AA})_{zz}$, we have

$$(\sigma_p^{AA})_{zz} = -[e^2 \hbar^2 / 2m^2 c^2 (\Delta E)] \langle r^{-3} \rangle_{2p} \times \sum_B [P_{xAxB} (2\delta_{AB} - P_{yAyB}) + P_{yAyB} (2\delta_{AB} - P_{xAxB}) + 2P_{xAyB} P_{yAxB}], \quad (5)$$

with corresponding expressions for the other diagonal components. Here $\sum_B$ goes over all atoms and $\langle r^{-3} \rangle_{2p}$ is the mean inverse cube of the distance from the nucleus for the carbon $2p$ atomic orbital. The $P_{\mu\nu}$ are elements of the charge-density and bond-order matrix in the unperturbed molecule; that is,

$$P_{\mu\nu} = 2 \sum_i^{\text{occ}} c_{i\mu} c_{i\nu}, \quad (6)$$

the summation being over all occupied molecular orbitals. The subscripts $x_A, x_B, y_A, y_B, z_A, z_B$ correspond to the various $2p$ atomic orbitals of atoms A and B, all referred to equivalently oriented coordinate systems. Since the usual solution NMR measurement is a rotational average, the mean value of the tensor is of interest here (though anisotropies can also be determined under appropriate conditions⁸). This quantity, σ_p^{AA} , has the form

$$\sigma_p^{AA} = -[e^2 \hbar^2 / 2m^2 c^2 (\Delta E)] \langle r^{-3} \rangle_{2p} \sum_B Q_{AB}, \quad (7)$$

where

$$Q_{AB} = \frac{4}{3} \delta_{AB} (P_{xAxB} + P_{yAyB} + P_{zAzB}) - \frac{2}{3} (P_{yAyB} P_{zAzB} + P_{zAzB} P_{xAxB} + P_{xAxB} P_{yAyB}) + \frac{2}{3} (P_{yAzB} P_{zAyB} + P_{zAyB} P_{xAzB} + P_{xAzB} P_{yAzB}). \quad (8)$$

A number of approximations are involved in Eqs. (5) and (7), the principal ones being the neglect of all two-center integrals and the replacement of the electronic excitation energies by a mean value ΔE .^{6,7}

To apply these general formulas to a carbon atom A with sp^2 hybridization in a planar conjugated system, we let B, C, and D be three neighboring carbon atoms to which A is bonded. All bond angles are taken to be 120° , although significant deviations do occur. If axes are chosen as in Fig. 1, the bond orders P_{zAzB} , P_{zAzC} , and P_{zAzD} are the mobile bond orders for the π -electron system as given by the Hückel molecular orbital approximation (without overlap).⁹ For the σ electron we assume localized bond orbitals of the form

$$\Psi(\sigma \text{ bond}, AB) = 2^{-1/2} [\phi(A; sp^2) + \phi(B; sp^2)], \quad (9)$$

where $\phi(A; sp^2)$ and $\phi(B; sp^2)$ are sp^2 hybrids on atoms A and B along the bond direction. Overlap is neglected in the normalization of Eq. (9). With these LCAO coefficients, Eq. (7) for the shielding constant σ_p^{AA} becomes

$$\sigma_p^{AA} = -[e^2 \hbar^2 / 2m^2 c^2 (\Delta E)] \langle r^{-3} \rangle_{2p} \times [2 + \frac{4}{3} (P_{zAzB} + P_{zAzC} + P_{zAzD})]. \quad (10)$$

If A is bonded to two carbon atoms, B and C, and one hydrogen atom D, a modified expression is applicable. Its exact form depends on the bond polarity. For a nonpolar AH bond, the σ -bonding orbital can be written

$$\Psi(\sigma \text{ bond}, AH) = 2^{-1/2} [\phi(A; sp^2) + \phi(H; 1s)], \quad (11)$$

where $\phi(H; 1s)$ is the hydrogen $1s$ atomic orbital. Substitution into Eq. (7) yields an expression which is identical to Eq. (10) except that the term P_{zAzD} is

⁵ A. Saika and C. P. Slichter, *J. Chem. Phys.* **22**, 26 (1954).

⁶ M. Karplus and T. P. Das, *J. Chem. Phys.* **34**, 1683 (1961).

⁷ J. A. Pople, *J. Chem. Phys.* **37**, 53, 60 (1962).

⁸ See, for example, A. D. Buckingham and E. G. Lovering, *Trans. Faraday Soc.* **58**, 2077 (1962).

⁹ A. Streitwieser, *Molecular Orbital Theory for Organic Chemists* (John Wiley & Sons, Inc., New York, 1962).

omitted. If, however, we wish to account for the possibility of some ionic character in the AH bond,⁶ we can express the σ orbital in the more general form

$$\Psi(\sigma \text{ bond, AH}) = 2^{-1/2}[(1 + \lambda_H)^{1/2}\phi(A; sp^2) + (1 - \lambda_H)^{1/2}\phi(H; 1s)], \quad (12)$$

where λ_H is the polarity parameter, which is probably small,¹⁰ but not necessarily zero. The only effect of a nonzero λ_H is to change $P_{z_A z_A}$ from 1 to $(1 + 2/3\lambda_H)$. Introducing this value for $P_{z_A z_A}$ into Eq. (7) for a carbon atom with 2C and 1H neighbors, we obtain

$$\sigma_p^{AA} = -[e^2\hbar^2/2m^2c^2(\Delta E)]\langle r^{-3} \rangle_{2p} \times [2 + \frac{4}{9}\lambda_H(1 - P_{z_A z_A}) + \frac{4}{9}(P_{z_A z_B} + P_{z_A z_C})]. \quad (13)$$

Corresponding terms could be introduced in a straightforward manner to account for the presence of other polar bonds (e.g., C-N), for polarity in C-C bonds introduced by electronegative or electropositive substituents, or for π - σ bond polarization. Although such effects may be of importance in the appropriate molecules, we shall not discuss them further in this paper.

Eqs. (10) and (13) involve the total mobile bond order for bonds from one particular atom. The difference between this and its highest possible value ($\sqrt{3}$) is often referred to as the free-valence index F for the atom.⁹ Thus Eqs. (10) and (13) may be written, respectively,

$$\sigma_p^{AA} = -[e^2\hbar^2/2m^2c^2(\Delta E)]\langle r^{-3} \rangle_{2p}[2 + \frac{4}{9}(\sqrt{3} - F_A)], \quad (14)$$

and

$$\sigma_p^{AA} = -[e^2\hbar^2/2m^2c^2(\Delta E)]\langle r^{-3} \rangle_{2p} \times [2 + \frac{4}{9}\lambda_H(1 - P_{z_A z_A}) + \frac{4}{9}(\sqrt{3} - F_A)]. \quad (15)$$

3. DISCUSSION AND APPLICATION

The general expressions, Eqs. (14) and (15), for the local paramagnetic contribution to the shielding constant show an explicit dependence on the pi-electron charge density ρ_A only through the polarity term λ_H . However, it is evident that three additional features of the electronic environment are important in determining the shielding. They are the mean excitation energy for the molecule, the dimension of the $2p$ atomic orbital, and the free-valence index. We shall consider each of these features separately and determine whether they are expected to contain any implicit dependence on ρ_A .

A. Electronic Excitation Energies

The excitation energies which have been replaced by a mean value ΔE in the derivation of Eqs. (14) and (15) are mainly for excited states involving

($\sigma \leftrightarrow \pi$) transitions, as well as for transitions within the sigma framework. The ($\pi \leftrightarrow \pi$) transitions which usually have the lowest excitation energies in conjugated hydrocarbons do not contribute. The quantity ΔE is, therefore, taken to be approximately equal to 10 eV as in earlier calculations on nonconjugated molecules.⁷

Since ΔE is a mean value for the whole molecule, any effects of excitation energies on relative shielding for different atoms are not taken into account in Eqs. (14) and (15). If the lower excited states could be represented as localized excitations on particular atoms, those atoms would presumably have a larger paramagnetic term σ_p^{AA} . For conjugated and aromatic carbon compounds, there do not seem to be any grounds for singling out individual atoms in this way, although certain substituents may lead to local excitations.

B. Atomic Orbital Dimensions

The factor $\langle r^{-3} \rangle_{2p}$ in Eqs. (14) and (15) leads directly to a correlation with electron density. This arises from the fact that the $2p$ orbitals involved are expected to expand as electrons are added to the atom. The resulting decrease in $\langle r^{-3} \rangle_{2p}$ will diminish the magnitude of the paramagnetic contribution σ_p^{AA} given by Eqs. (14) and (15), and thereby increase (algebraically) the total shielding. The magnitude of the effect, which has also been considered in the evaluation of quadrupole coupling constants,¹¹ may be estimated by using Slater atomic $2p$ orbitals to calculate $\langle r^{-3} \rangle_{2p}$ and by choosing the nuclear charge Z_A according to Slater's rules.¹² Since these yield a Z_A value of 3.25 for a neutral carbon atom and a screening per $2p$ electron of 0.35, we can write

$$Z_A = 3.25 - 0.35(\rho_A - 1), \quad (16)$$

where ρ_A is the local pi-electron density on atom A. This assumes that the Slater's rules, which were originally formulated for integral numbers of electrons, may reasonably be extended to atoms with fractional electron populations. For a Slater $2p$ orbital, $\langle r^{-3} \rangle_{2p}$ is found to be

$$\langle r^{-3} \rangle_{2p} = \frac{1}{24}(Z_A/a_0)^3, \quad (17)$$

where a_0 is the Bohr radius. Substituting from Eq. (16) and (17) into Eq. (14), we have

$$\sigma_p^{AA} = -[e^2\hbar^2/48a_0^3m^2c^2(\Delta E)] \times [3.25 - 0.35(\rho_A - 1)]^3[2 + \frac{4}{9}(\sqrt{3} - F_A)] \quad (18)$$

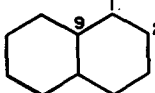
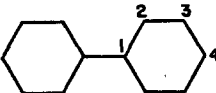
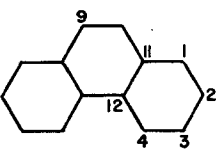
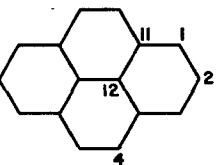
and a corresponding result from Eq. (15). It is clear that a significant ρ_A dependence is introduced by the $\langle r^{-3} \rangle_{2p}$ term (see below).

¹⁰ C. A. Coulson, *Valence* (Oxford University Press, New York, 1952).

¹¹ C. H. Townes and A. L. Schalow, *Microwave Spectroscopy* (McGraw-Hill Book Company, Inc., New York, 1955), p. 238.

¹² J. C. Slater, *Phys. Rev.* **36**, 57 (1930).

TABLE I. Chemical shifts for some alternant hydrocarbons.

	Free valence ^a	Chemical shift ^b	
		$\Delta\sigma(\text{calc})^c$	$\Delta\sigma(\text{exp})^d$
Naphthalene			
	$F_1=0.452$	$\Delta\sigma_1=+2.4$	$\Delta\sigma_1=+0.8$
	$F_2=0.404$	$\Delta\sigma_2=+0.23$	$\Delta\sigma_2=+0.8$
	$F_9=0.104$	$\Delta\sigma_9=-13.6$	$\Delta\sigma_9=-8.4$
Biphenyl			
	$F_1=0.124$	$\Delta\sigma_1=-12.7$	$\Delta\sigma_1=-9.9$
	$F_2=0.436$	$\Delta\sigma_2=+1.7$	$\Delta\sigma_2=-0.4$
	$F_3=0.395$	$\Delta\sigma_3=-0.18$	$\Delta\sigma_3=-0.4$
	$F_4=0.412$	$\Delta\sigma_4=+0.60$	$\Delta\sigma_4=-0.4$
Phenanthrene			
	$F_1=0.450$	$\Delta\sigma_1=+2.3$	$\Delta\sigma_1=+0.6$
	$F_2=0.402$	$\Delta\sigma_2=+0.14$	$\Delta\sigma_2=+0.6$
	$F_3=0.407$	$\Delta\sigma_3=+0.37$	$\Delta\sigma_3=+0.6$
	$F_4=0.440$	$\Delta\sigma_4=+1.9$	$\Delta\sigma_4=+0.6$
	$F_9=0.451$	$\Delta\sigma_9=+2.4$	$\Delta\sigma_9=+0.6$
	$F_{11}=0.108$	$\Delta\sigma_{11}=-13.4$	$\Delta\sigma_{11}=-4.8$
	$F_{12}=0.139$	$\Delta\sigma_{12}=-12.0$	$\Delta\sigma_{12}=-4.8$
Pyrene			
	$F_1=0.468$	$\Delta\sigma_1=+3.2$	$\Delta\sigma_1=+1.0$
	$F_2=0.393$	$\Delta\sigma_2=-0.28$	$\Delta\sigma_2=+1.0$
	$F_4=0.452$	$\Delta\sigma_4=+2.4$	$\Delta\sigma_4=+1.0$
	$F_{11}=0.110$	$\Delta\sigma_{11}=-13.3$	$\Delta\sigma_{11}=-4.4$
	$F_{12}=0.148$	$\Delta\sigma_{12}=-11.5$	$\Delta\sigma_{12}=-8.1$

^a From *Dictionary of Values of Molecular Constants*, edited under the direction of the Mathematical Institute, Oxford, and the Centre de Chimie Theorique de France, Paris, 1955, Vol. II.

^b Parts per million with reference to benzene.

^c Calculated from Eq. (19) with the assumption that $(\rho_A-1)=0$.

^d From reference 2. Values bracketed together correspond to unresolved results presented in the paper. It should also be noted that some of the assignments are tentative.

C. Free Valence

The free-valence contribution in Eqs. (14) and (15) arises from the mobil bond order and clearly does not correlate with the electron density. In the series $C_7H_7^+$, C_6H_6 , $C_5H_5^-$, for example, the free valencies are 0.448, 0.399, and 0.438, respectively, and give a positive contribution to the chemical shift for both ions relative to benzene.

D. Application

To estimate the magnitudes of the various effects appearing in the theoretical equations, we apply them

to chemical shifts relative to benzene as the standard. Further, it is helpful to assume that the charge densities are close to unity, that λ_H is close to zero, and that the free valencies are close to the benzene value of 0.399. With these limitations, the general formulas can be linearized. We have

$$\Delta\sigma_A(\text{ppm}) = 10^6(\sigma_A - \sigma_{\text{benzene}}) \\ \cong (86.7 + 46.0\lambda_H)(\rho_A - 1) + 46.0(F_A - 0.399), \quad (19)$$

where λ_H equals zero for a carbon atom bonded to three other carbons.

From Eq. (19) it is clear that the direct size effect on the atomic wavefunction provides a dependence of $\Delta\sigma_A$ on the charge ρ_A that is in the correct direction and of the right magnitude. The actual numerical value that appears in Eq. (19) is about half that observed experimentally [Eq. (1)], but in view of the many approximations involved, only the order of magnitude is likely to be significant.

Equation (19) shows also that the charge dependence may be affected by the atoms bonded to the one under consideration; that is, for a carbon bonded to three other carbons ($\lambda_H=0$) the coefficient of (ρ_A-1) would be somewhat different from a carbon bonded to two others and one hydrogen. However, since λ_H is small (probably $0 \leq \lambda_H < 0.2$), a careful search for the polarity effect would seem to be required [but, see also the effects mentioned after Eq. (13)].

The free-valence dependence of the shielding given in the last term of Eq. (19) will often be dominated by the charge effect. For the series $C_7H_7^+$, C_6H_6 , $C_5H_5^-$ mentioned above, the free-valence contribution calculated from Eq. (19) is approximately +2 ppm, while a shift on the order of ± 15 ppm is computed from the ρ_A values. Thus, to isolate the free-valence term, it would be desirable to examine compounds in which the local densities are approximately constant. For pi-electron systems, the usual theories suggest that this condition is best satisfied for alternant hydrocarbons. The only data presently available in the literature are some preliminary analyses of Lauterbur.² Table I gives comparison of the experimental results with calculations employing Hückel theory free valencies in Eq. (19), subject to the assumption that $(\rho_A-1)=0$ for all

TABLE II. Chemical shift results for 1,3-butadiene.

Atoms	Free valence	Chemical shift ^a	
		$\Delta\sigma(\text{calc})^b$	$\Delta\sigma(\text{exptl})^c$
2, 3	0.391	-0.4	-9.5
1, 4	0.838	+20.2	+11.5

^a Part per million relative to benzene.

^b From Eq. (19) with the assumption that $(\rho_A-1)=0$.

^c From reference 3.

positions. The general trends (e.g., the difference between the 1, 2, and the 9 position in naphthalene) are given correctly. As to the finer details of the comparison, a more complete analysis of the spectra is required to make any meaningful statements.^{12a} Also, of course, the charge is not exactly unity in all positions, and some correction might be introduced on the basis of improved molecular orbital treatments for the pi electrons.¹³

A corresponding application of the theory can be

^{12a} Note added in proof: We understand from W. G. Schneider that, in unpublished work with H. Spiesscke, he has found the separate chemical shifts for the 1 and 2 positions in naphthalene. The results are: $\Delta\sigma_1 = +2.0$ and $\Delta\sigma_2 = +0.1$ ppm relative to benzene, in good agreement with the calculations quoted in Table I.

¹³ See, for example, the naphthalene calculation of H. O. Pritchard and F. H. Skinner, *Trans. Faraday Soc.* **50**, 457 (1955).

made to 1,3-butadiene which has been studied experimentally by Spiesscke and Schneider.³ The results are summarized in Table II. The separation between the two butadiene positions is given reasonably well, but not the positions relative to benzene. However, this comparison is complicated by the neglect of several factors, including the unequal bond lengths in butadiene.

It is clear from these results that Eq. (19) is of the correct form to explain some of the observed empirical trends. Hopefully, a fuller test and additional applications of Eq. (19) and its obvious extensions will be provided by future studies of appropriate molecules.

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Structure of Hexagonal Silver Iodide*

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(Received 28 December 1962)

A detailed three-dimensional refinement of the structure of hexagonal silver iodide from room-temperature photographic data showed very little deviation from the ideal wurtzite-type structure. Using anisotropic temperature factors a final R value of 6.6% was obtained. The space group is $P6_3mc$, $a = 4.592$, $c = 7.510$, $c/a = 1.635$, $\mu = 0.374$. A general linear relation of the position parameter μ to the c/a axial ratio was found to apply to compounds having this structure type.

INTRODUCTION

AS part of a study of the crystallographic properties of silver iodide, the structure of the hexagonal wurtzite-type phase of this compound has been redetermined. Silver iodide has been reported in four different modifications. At temperatures below 147° it is dimorphous, with wurtzite- and zinc-blende-type structures.¹⁻³ These obey the Grimm-Sommerfeld rule, with a 4:1 valency electron to atom ratio. Between 147°C and the melting point a disordered body-centered cubic structure is stable,⁴ and a high-pressure NaCl-type structure exists above about 3000 bars.⁵

The ideal wurtzite structure, composed of regular

tetrahedra, has a c/a axial ratio of 1.633. Deviations from this ideal value are possible and have been observed for most structures of this type. It has been noted that when the compound is dimorphous and exists also in the zincblende structure the c/a ratio is close to the ideal value. Regardless of its numerical value the bond distribution need not be perfectly tetrahedral. In order to study any distortion in terms of the structure it is necessary to determine a positional parameter μ , which represents the interatomic distance of two bonded atoms in the direction of the hexagonal axis c , expressed as a fraction of the length of c .

Silver iodide was first reported as analogous to zinc oxide by Aminoff⁶ on the basis of powder and Laue x-ray diffraction photographs. Kolkmeijer and van Hengel,³ using powder-diffraction data alone, reported the value of μ as 0.371. The agreement between calculated and observed intensities was not very good, and high order maxima were neglected completely. They attributed the discrepancy to preferred orien-

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† This work constituted a portion of a thesis submitted to Georgetown University, Washington, D. C., in partial fulfillment of the requirements for the Ph.D. degree (1962).

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⁶ G. Aminoff, *Z. Krist.* **57**, 180 (1922).